Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *CRISISLTLSUM* | ***Context:*** *Mexico City and Central America Bilingual Crisis Journalism*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark evaluates only English-language output: timeline-extraction accuracy is measured against expert annotations on English tweets [Q91, Q94, Q96]; timeline summaries are evaluated using SacreROUGE [Q98, Q100] against English-language reference summaries written by Anglophone MTurk workers [Q99]; the human-evaluation rubric is written in English and scores coherence against English-language writing norms [Q106, Q153]. The target deployment requires bilingual output (Mexican Spanish + American English) with the elicitation specifying that English summaries should be more directly publication-ready and Spanish summaries adapted by journalists (A4). The benchmark provides no Spanish-language reference summaries, no Spanish coherence rubric, no bilingual evaluation protocol, no AP Español alignment metric, and no register-appropriateness scoring. The web research further documents that ROUGE correlates poorly with human judgments for Spanish summarization, with BERTScore-precision (multilingual) and LLM-as-judge being current best practice [WEB-17, WEB-18] — none of which the benchmark uses. The mismatch with the deployment's bilingual output requirement is total for Spanish.

ID	Evidence
Q91	“For evaluation, we use the average timeline-level accuracy.” (p.7)
Q94	“For timeline extraction results on the TEST set, we compare both the human-level performance and the best model's performance against experts' annotations.” (p.8)
Q96	“The best timeline extraction model achieves 73.51 average timeline accuracy, whereas the human-level performance is 88.98.” (p.8)
Q98	“We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004).” (p.8)
Q99	“The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels.” (p.8)
Q100	“We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores.” (p.8)
Q106	“Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022).” (p.8)
Q153	“For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors.” (p.21)
WEB-17	Standard ROUGE variants correlate weakly with human judgments for Spanish summarization; BERTScore-precision with multilingual models is recommended (https://arxiv.org/pdf/2503.17039)
WEB-18	LLM-as-judge approaches show strong correlation with human judgments in multilingual summarization evaluation (https://aclanthology.org/2024.emnlp-main.1085.pdf)

Strengths

- The English-language output evaluation [Q98, Q106] is methodologically sound and directly applicable to the deployment's English-output evaluation needs, where English summaries are expected to be higher-quality and more publication-ready (elicitation A4).
- Multi-reference ROUGE [Q99, Q100] and the four-axis human evaluation [Q106] together provide a defensible English evaluation framework that can be reused for the English half of bilingual output.

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Q98	“We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004).” (p.8)
Q99	“The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels.” (p.8)
Q100	“We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores.” (p.8)
Q106	“Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022).” (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Partial match — output modality (free-form text summary) matches deployment requirements, but the language coverage (English-only) misses 50% of the bilingual requirement [Q99, Q106; elicitation A4].

OF-2: Not applicable — deployment does not require text-to-speech output (journalists work with text artifacts).

OF-3: Not applicable — target users are professional journalists with high literacy; no accessibility constraints alter the form requirement.

OF-4: Form mismatch: complete absence of Spanish-language summarization evaluation [Q99, Q106, Q153] directly contradicts the bilingual deployment requirement, severely harming external validity for the Spanish output channel.

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Q98	“We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004).” (p.8)
Q99	“The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels.” (p.8)
Q100	“We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores.” (p.8)
Q106	“Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022).” (p.8)
Q153	“For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors.” (p.21)
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Information Gaps

- Operationalizable AP Español rubric criteria not openly accessible
- No Spanish crisis-summarization reference dataset exists to enable bilingual evaluation construction

Requires Expert Verification

- AP Español editor input to construct a Spanish coherence/register rubric
- Mexican journalist evaluation panel to validate any Spanish-language summarization quality framework

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Remediation

Gap 1: No Spanish-language summarization evaluation; English-only ROUGE [Q99, Q100] and English coherence rubric [Q153]

Recommendation: Implement a bilingual evaluation protocol: BERTScore with multilingual backbone (XLM-R or mBERT) and LLM-as-judge for Spanish summaries [WEB-17, WEB-18], alongside human evaluation by Spanish-speaking journalists. Maintain the existing English ROUGE pipeline for the English output channel where it remains defensible.

ID	Evidence
Q99	“The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels.” (p.8)
Q100	“We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores.” (p.8)
Q153	“For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors.” (p.21)
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